

Cold joints in concrete with **D5** exhibit no leakage at pressures of up to 12 bar

Tests were conducted at the Belarusian National Technical University (Test report #302, dated 18.05.2022)

- To test the waterproofing of cold joints in concrete with the admixture D5, the cylindrical samples (150 mm in diameter and 150 mm high) were cast.
- The samples were made in 2 stages: first, a cylinder half (semi-cylinder) was cast (See Fig.). After 24 hours, the hardened half of the cylinder was removed from the mold, the vertical surface of the cold joint was cleaned from cement crust and moistened.
- The cylinder half was then placed back into the mold and cast with ready-mixed concrete with D5, creating a concrete cylinder with a vertical cold joint.
- Six cylindrical samples with cold joints were cast. They were cured in a standard curing room for 28 days, then subjected to a water permeability test (wet spot method) in accordance with GOST 12730.5-2018.
- The water pressure was increased in increments of 0.2 MPa every 1-2 minutes, with each stage maintained for 24 hours.
- **Testing demonstrated that cold joints in samples of concrete with D5 remained watertight under pressures of up to 12 bar (1.2 MPa).**

Cold joint

Холодный шов

